MILITARY SPECIFICATION

SHELL, TEST, FUZE TELEMETRY

1. SCOPE

1.1 This specification covers special requirements and provides the quality assurance provisions, and the preparation for delivery criteria for one type of Test Shell used on the XM454 project.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Pressure Sensitive, Adhesive, Waterproof - for Packaging and Sealing

MILITARY

MIL-B-20292 - Blanks, Rotating Band for Projectiles
MIL-B-20295 - Blanks, Centrifugal Cast, Rotating Band

STANDARDS

FEDERAL

Federal Test Method Standard No. 151

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

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DRAWINGS

ORDNANCE CORPS

75-14-371D - Grommets, for 155mm and 8 In. and 240mm, Projectiles Assembly, List of Parts and List of Spec, and STD
9203276 - Pin Straight Headless
9204730 - Fuze Telemetry Test Shell T5132
9204731 - Body
9204732 - Slug
9204733 - Windshield
9214158 - Base, Alternative
9214159 - Body, Alternative
ET-9204730 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test shell shall comply with all requirements specified on Drawing (dwg.) 9204730, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Rotating band (see dwgs. 9204731 and 9214158).

3.5.1 Blanks. The rotating band blanks shall be in the form of a solid ring, annealed if necessary, to comply with specification MIL-B-20292 or MIL-B-20295 when tested as specified in 4.4.1.1.

3.5.2 Physical properties. After removal from body, a 2 inch gage test coupon shall show the following physical properties: Tensile strength 30,000 pounds per square inch minimum. Elongation of 25 percent when tested as specified in 4.4.1.2.

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3.5.3 Seating. After seating of the band before machining, the diametrical clearance between the band and the seat shall not exceed 0.005 inch at any point of measurement on the band when computed as specified in 4.4.1.2.

3.5.4 Damaged rotating band. Each rotating band on each projectile assembly shall be inspected for the following defects.

a. Cracks

b. Any dent, nick, scratch or gouge which exceeds the allowable depth (see Fig. I).

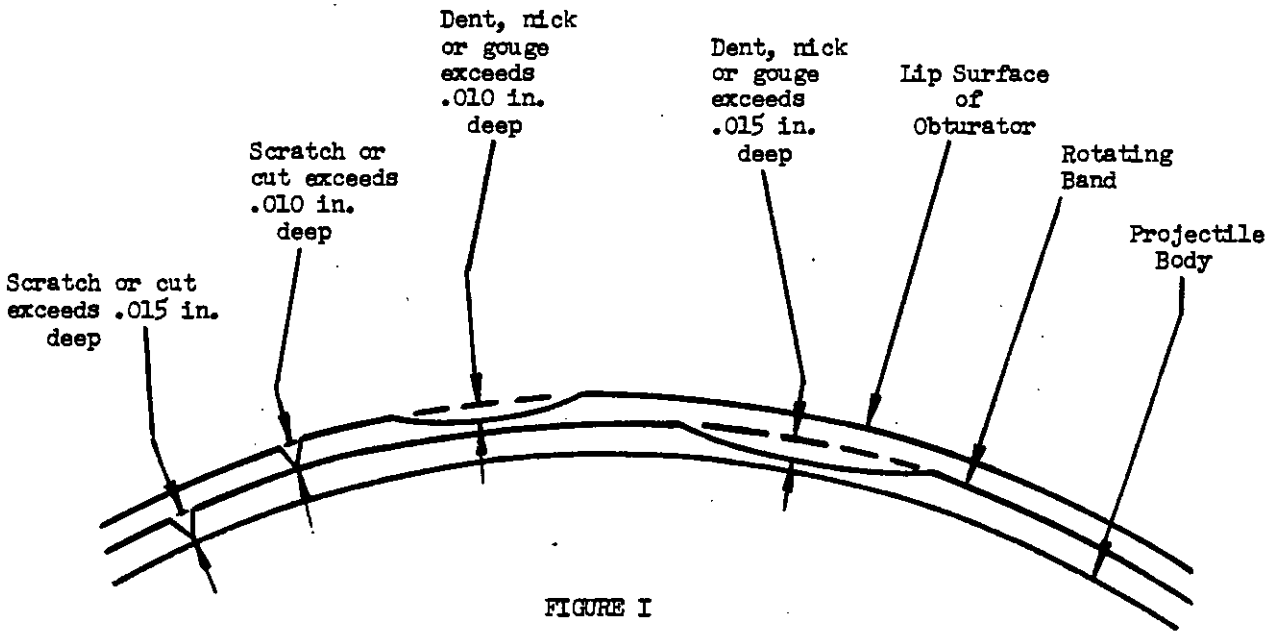


FIGURE I

ROTATING BAND INSPECTION CRITERIA

c. Foreign matter embedded in rotating band material.

Any rotating band which has any of the defects listed in a, b and c shall be rejected and removed from the lot when inspected as specified in 4.4.1.3.

3.6 Metal to metal contact (see dwg. 9204730).

3.6.1 Metal to metal contact between body and slug. Metal to metal contact between body and slug shall be assured when the calculated dimension (see 4.4.2.1) equals the actual dimension measured after assembly of the rear pins.

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3.6.2 Metal to metal contact between body and windshield. Metal to metal contact between the body and windshield shall be examined as specified in 4.4.2.2. At least 50% of the surface between the body and windshield shall be in contact.

3.6.3 Metal to metal contact between alternative base and alternative body. When alternative base (see dwg. 9214158) and alternative body (see dwg. 9214159) are used in the assembly, metal to metal will be considered established when the gap in the joint is calculated in accordance with the drawing. The gap shall not exceed .001 inch or show a variation of more than .0005 inch when measured as specified in 4.4.2.3.

3.7 Interference fit of pins (see dwg. 9204730). Pin holes which are machined to a diameter larger than nominal (.3750 + .0003 or .4990 + .0005 as applicable) in accordance with notes on dwg. 9204730, shall be uniform diameter within the limits of the nominal tolerance (.0003 or .0005 inch as applicable) for the full depth. Each such hole and its mating pin shall be marked in a manner to permit ready identification of the proper hole and pin combination until the assembly is performed.

3.8 Paint adhesion. Painted items shall show satisfactory paint adhesion when tested as prescribed in 4.4.3. Unsatisfactory adhesion shall be indicated by any of the following conditions of exposure of bare metal or underlying pretreatment.

a. Any spot exceeding 1/8 inch average diameter.
b. More than one spot exceeding 1/16 inch average diameter.
c. More than 5 spots any diameter.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg. 9204730), and 2 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Table I

First article inspections and tests

	Examination or test	Contractor's inspection permissable before submission	Parts	Assemblies
1.	Classification of characteristics	X	X	X
2.	Rotating band (see 3.5)	X		X
3.	Metal to metal contact (see 3.6)	X		X
4.	Pin and hole rework procedures (see 3.7)	X		X
5.	Charpy (see 4.3.3.2)	X	X	
6.	Tensile (see 4.3.3.3)	X	X	
7.	Shear (see 4.3.3.4)	X	X	
8.	Paint adhesion (see 3.8)	X		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

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4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table II

Lot size	Sample size
1- 155	100% inspection
156- 200	155
201- 250	175
251- 300	190
301- 350	200
351- 400	210
401- 450	220
451- 500	225
501- 550	230
551- 600	235
601- 650	240
651- 700	243
701- 750	247
751- 800	250
801- 900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Minor characteristic table III

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

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4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Slug (see dwg. 9204732 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Outside diameter, min.	Gage
102.	Inside diameter	Gage
103.	Concentricity of inside diameter	Gage
104.	Total length	Gage
105.	Perpendicularity of rear and forward faces	Gage
106.	Chamfer	Visual
107.	Materials (see 3.1)	Visual
Minor:		
201.	Surface finish	Visual
202.	Stamping, metal not raised by stamping	Visual
203.	Serialization (see 3.4)	Visual
204.	Protective finish present no bare spot	Visual
205.	Workmanship (see 3.9)	Visual

4.3.2.1.2 Windshield (see dwg. 9204733 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of thread	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of thread	Gage
104.	Chamfer on small inside diameter	Gage
105.	Concentricity of tapered outside diameter including perpendicularity with left shoulder	Gage
106.	Outside diameter to sharp corners, min.	Gage
107.	Left outside diameter, min.	Gage
108.	Left inside diameter, max.	Gage
109.	Total length	Gage
110.	Length from left outside shoulder to right outside shoulder	Gage
111.	Right outside shoulder parallel with left outside shoulder	Gage

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112. Depth of cavity left of thread.....Gage
113. Length from right face to right outside shoulder.....Gage
114. Length between large cavity and cavity left of thread.....Gage
115. Inside diameter left of thread.....Gage
116. Concentricity of inside diameter left of thread.....Gage
117. Outside diameter above thread clearance, max.....Gage
118. Length of outside diameter above thread clearance, max.....Gage
119. Width of large slot in thread.....Gage
120. Large slot in thread symmetrical with right inside diameter.....Gage
121. Width of inside slot.....Gage
122. Inside slot symmetrical with right inside diameter.....Gage
123. Length from bottom of inside slot to circumference of small inside diameter.....Gage
124. Forward inside diameter.....Gage
125. Concentricity of right inside diameter.....Gage
126. Width of two small slots in thread, min.....Gage
127. Location of two small slots in thread.....Gage
128. Small inside diameter.....Gage
129. Steps present on surfaces noted.....Visual/Manual
130. Material (see 3.1).....Visual

Minor:
201. Radius or chamfer present.....Visual
202. Surface finish.....Visual
203. Stamping, metal raised by stamping removed.....Visual
204. Serialization (see 3.4).....Visual
205. Protective finish present, no bare spot.....Visual
206. Workmanship (see 3.9).....Visual

4.3.2.1.3 Body (see dwg. 9204731) or base (see dwg. 9204158) covering a detail of dwg. 9204730) after finish machining rotating band seat and rough machining.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Width of rotating band groove.....Gage
102. Small diameter of rotating band groove.....Gage

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103. Projections, not missing, damaged or incorrect number on bottom of rotating band groove.....Visual
104. Notches in projections on bottom of rotating band groove not missing or incorrectly spaced.....Visual

Minor:
201. Workmanship (see 3.9).....Visual

4.3.2.1.4 Body (see dwg. 9204731 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Bourrelet diameters.....Gage
102. Left cavity diameter, max.....Gage
103. Diameter of thread relief, min.....Gage
104. Diameter left of smallest inside diameter.....Gage
105. Smallest inside diameter.....Gage
106. Concentricity of smallest inside diameter.....Gage
107. Depth of left thread cavity.....Gage
108. Depth to bottom of thread relief.....Gage
109. Depth to top of thread relief, min.....Gage
110. Pitch diameter of large thread.....Gage
111. Minor diameter of large thread.....Gage
112. Concentricity of large thread.....Gage
113. Length of bourrelet diameter right of rotating band seat, min.....Gage
114. Pitch diameter of 3 threaded holes.....Gage
115. Minor diameter of 3 threaded holes.....Gage
116. Length of right bourrelet diameter, min.....Gage
117. Outside diameter between bourrelets, min.....Gage
118. Right outside diameter, max.....Gage
119. Diameter of right cavity, max.....Gage
120. Concentricity of right inside diameter.....Gage
121. Perpendicularity of right face.....Gage
122. Depth of right cavity.....Gage
123. Perpendicularity of inside shoulder.....Gage
124. Total depth of cavities, max.....Gage
125. Total length of rotating band.....Gage
126. Diameter of small grooves in rotating band.....Gage
127. Diameter of large groove in rotating band.....Gage
128. Large outside diameter of rotating band.....Gage
129. Total length of body.....Gage
130. Right inside chamfer present.....Visual

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Minor:
201. Surface finish.....Visual
202. Stamping, metal raised by stamping removed.....Visual
203. Serialization (see 3.4).....Visual
204. Protective finish present, no bare spot.....Visual
205. Workmanship (see 3.9).....Visual

4.3.2.1.5 Base, alternative (see dwg. 9214158 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Bourrelet diameters.....Gage
102. Pitch diameter of thread.....Gage
103. Minor diameter of thread.....Gage
104. Concentricity of thread.....Gage
105. Length to rotating band groove.....Gage
106. Total length.....Gage
107. Length of right outside diameter.....Gage
108. Smallest inside diameter.....Gage
109. Concentricity of smallest inside diameter.....Gage
110. Diameter of left cavity, min.....Gage
111. Diameter of thread clearance, min.....Gage
112. Diameter of middle cavity, max.....Gage
113. Concentricity of middle cavity.....Gage
114. Diameter of right cavity, max.....Gage
115. Concentricity of right cavity.....Gage
116. Right outside diameter.....Gage
117. Concentricity of right outside diameter.....Gage
118. Radius missing.....Visual
119. Depth of right cavity.....Gage
120. Depth of middle cavity.....Gage
121. Surface finish 63.....Visual
122. Perpendicularity of right inside shoulder.....Gage
123. Depth of left cavity.....Gage
124. Depth of thread clearance.....Gage
125. Depth of thread.....Gage
126. Length of rotating band.....Gage
127. Diameter of small grooves in rotating band.....Gage
128. Diameter of large groove in rotating band.....Gage
129. Large outside diameter of rotating band.....Gage
130. Forward inside chamfer missing.....Visual
131. Material (see 3.1).....Visual

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Minor:

201.	Surface finish	Visual
202.	Stamping, metal raised by stamping removed	Visual
203.	Serialization (see 3.4)	Visual
204.	Protective finish present, no bare spot	Visual
205.	Workmanship (see 3.9)	Visual

4.3.2.1.6 Body alternative (see dwg. 9214159 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101.	Diameter left of bourrelet, min	Gage
102.	Concentricity of diameter left of bourrelet	Gage
103.	Left outside diameter, min	Gage
104.	Surface finish 63	Visual
105.	Chamfers present	Visual
106.	Length of left outside diameter	Gage
107.	Location of threaded hole	Gage
108.	Perpendicularity of left shoulder	Gage
109.	Pitch diameter of threaded holes	Gage
110.	Minor diameter of threaded holes	Gage
111.	Total length	Gage
112.	Length from end of bourrelet to right face	Gage
113.	Perpendicularity of right face	Gage
114.	Bourrelet diameter	Gage
115.	Concentricity of bourrelet diameter	Gage
116.	Diameter to sharp corner right face	Gage
117.	Large inside diameter, max	Gage
118.	Concentricity of large inside diameter	Gage
119.	Small inside diameter, max	Gage
120.	Concentricity of small inside diameter	Gage
121.	Length of large inside diameter	Gage
122.	Perpendicularity of inside shoulder	Gage
123.	Radius missing	Visual
124.	Material (see 3.1)	Visual

Minor:

201.	Surface finish	Visual
202.	Stamping, metal raised by stamping removed	Visual
203.	Serialization (see 3.4)	Visual
204.	Protective finish present, no bare spot	Visual
205.	Workmanship (see 3.9)	Visual

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4.3.2.1.7 Grommets for 155MM, 6 in. and 240MM Projectiles, Assembly, list of parts and list of spec and std (see dwg. 75-14-271 covering a detail of dwg. 9204730).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Staples flat against liner	Visual
102.	No part missing or incorrectly assembled	Visual
Minor:		
201.	Protective finish present, no bare spot	Visual
202.	Paint present no bare spot	Visual
203.	Workmanship (see 3.9)	Visual

4.3.2.1.8 Fuze telemetry test shell T5132 (see dwg. 9204730), Assembly of slug before pin assembly and assembly of windshield.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter of 3 pin holes	Gage
102.	Finish on pin holes	Visual
103.	Interference fit of pins (see 3.7)	Gage
Minor:		
201.	Workmanship (see 3.9)	Visual

4.3.2.1.9 Fuze telemetry test shell T5132 (see dwg. 9204730) before assembly of windshield and after assembly of pins.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Pins not above flush and not more than .020 inch below flush	Gage
102.	Proper metal to metal contact (see 3.6.1) between slug and body	Gage
103.	Pitch diameter on threaded hole	Gage
104.	Minor diameter on threaded hole	Gage

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105. Large diameter in threaded hole, min.....Gage
106. Small diameter in threaded hole.....Gage
107. Small diameter in threaded hole symmetrical with inside slot.....Gage
108. Depth of large hole in threaded hole.....Gage
109. Depth of thread in threaded hole, min.....Gage
110. Staking of pins.....Visual

Minor:
201. Workmanship (see 3.9).....Visual

4.3.2.1.10 Fuze telemetry test shell T5132 (see dwg. 9204730) before assembly of pins and after assembly of windshield.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of pin holes.....Gage
102. Finish on pin holes.....Visual
103. Interference fit of pins (see 3.7).....Gage

Minor:
201. Workmanship (see 3.9).....Visual

4.3.2.1.11 Fuze telemetry test shell T5132 (see dwg. 9204730).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Pins not above flush and not more than .020 inch below flush.....Gage
102. Metal to metal contact (see 3.6.2) between windshield and body.....Gage
103. Staked pins.....Visual
104. No damage to rotating band before assembly of grommet (see 3.5.4).....Visual
105. Grommet properly assembled.....Visual/Manual

Minor:
201. Paint present no bare spot.....Visual
202. Marking.....Visual

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203. Serialization (see 3.4).....Visual
204. Workmanship (see 3.9).....Visual

4.3.2.1.12 Fuze telemetry test shell T5132 (see dwg. 9204730) alternate body construction before pin assembly and after assembly of alternative body.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of pin holes.....Gage
102. Finish on pin holes.....Visual
103. Interference fit of pins (see 3.7).....Gage

Minor:
201. Workmanship (see 3.9).....Visual

4.3.2.1.13 Fuze telemetry test shell T5132 (see dwg. 9204730 alternative body construction).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Improper metal to metal contact (see 3.6.3).....Gage
102. Pins not above flush and not more than .015 inch below flush.....Gage

Minor:
201. Workmanship (see 3.9).....Visual

4.3.2.1.14 Pin, straight, headless (see dwg. 9203276 covering a detail of dwg. 9204730). This paragraph applies when the alternate body construction is used.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Length.....Gage
102. Chamfer left end.....Gage

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103. Chamfer right end.....Gage
104. Surface finish.....Gage
105. Rockwell hardness.....Gage
106. Diameter.....Gage

Minor:
201. Toolmakers ink present.....Visual
202. Workmanship (see 3.9).....Visual

4.3.2.1.15 Preparation for delivery (see section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual
Minor: None defined.		

4.3.3 Testing. All the tests specified shall be performed. Tests do not have to be performed in the order specified.

	Tests	Test Classification
a.	Rotating band	Major
b.	Charpy	Major
c.	Tensile	Major
d.	Shear	Major
e.	Paint Adhesion	Minor

Test a is considered destructive when sample parts are used and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Rotating band (see 3.5).

4.3.3.1.1 Blanks. Two samples shall be selected at random from each lot of blanks and tested as specified in 4.4.1.1. If any sample fails to comply with the requirement 3.5.1, the lot of blanks shall be rejected.

4.3.3.1.2 Physical properties and band seating. Prior to machining the rotating bands in the lot, a sample projectile shall be selected at random from each lot of 50 or less produced or the dummy sample representing the lot shall be tested in accordance with 4.4.1.2. If the sample fails to comply with the requirements 3.5.2 and 3.5.3 the lot of banded projectiles shall be rejected.

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4.3.3.1.3 Damaged rotating band (see 3.5.4). Each projectile shall be examined for rotating band damage in accordance with 4.4.1.3. If any band fails to comply with requirement 3.5.4, the assembly shall be rejected and removed from the lot.

4.3.3.2 Charpy.

4.3.3.2.1 Charpy (see dwgs. 9204731, 9204732, 9204733, 9214158 and 9214159). A sample shall be selected at random from each heat-treated lot, prepared and tested in accordance with 4.4.4.1. If any sample fails to comply with drawing requirements, the production lot or lot of material shall be rejected.

4.3.3.2.2 Charpy (see dwg. 9203276). A sample shall be prepared and tested in accordance with 4.4.4.2. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.3 Tensile.

4.3.3.3.1 Tensile (see dwgs. 9204731 and 9204732). A sample shall be selected from random from each heat-treated lot, prepared and tested in accordance with 4.4.4.1. If any sample fails to comply with drawing requirements, the production lot or lot of material shall be rejected.

4.3.3.3.2 Tensile (see dwgs . 9214158 and 9214159). A sample shall be selected from random from each heat-treated lot, prepared and tested in accordance with 4.4.4.1. If any sample fails to comply with drawing requirements, the production lot or lot of material shall be rejected.

4.3.3.4 Shear (see dwg. 9203276). One sample shall be selected at random from each heat-treated lot and tested as specified in 4.4.5. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.5 Paint adhesion (see 3.8). Each shell shall be tested as specified in 4.4.3. If any shell fails to comply with 3.8, the shell shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-9204730. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.2 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Rotating band (see dwgs. 9204731 and 9214158).

4.4.1.1 Blanks (see 3.5.1). The rotating band blanks shall be tested in accordance with Specification MIL-B-20295 or MIL-B-20292.

4.4.1.2 Band seating and physical properties (see 3.5.2 and 3.5.3). Prior to removal the unmachined band shall be measured on three diameters 120 degrees apart. At each of the three diameters, measurement along the width of the band shall be taken at stations 1, 2 and 3 totaling 9 measurements in all. Stations 1 and 3 are 5/16 inch from the ends, and Station 2 is at the center. The location of the points of measurement shall be marked on the band along with the orientation of the band with the base. The band shall be removed in a manner that will keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 18 measurements) and the diameter of the band seat at these points shall also be measured. To each of the band seat diameters add the sum of the corresponding two band thicknesses and subtract from the corresponding diameter previously determined. The result will indicate twice the diametral clearance between the band and its seat as indicated in 3.5.3. If the band is well seated, a negative clearance or interference may be indicated. When irregular band profiles interfere with measurement, the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The diameter of the machined band shall exceed the bourrelet diameter. Since a piece must be destroyed to test clearances, a dummy component of similar material will be used for the test providing the dimensions of the band seat is within dwgs. 9204731 and 9214158 each time a band is tested. The machine settings used in banding the dummy component or scrap piece shall be the same as the lot being banded. One specimen shall be tested for every 50 pieces or less produced. A maximum of 6 cuts in the rotating band are permissible across the band seat in the test piece. The removed band shall be carefully flattened without hammering and test coupons shall be machined and tested in accordance with Federal Test Method Standard No. 151 except the overall length specified for the specimen shall be determined by the contractor. Band seating and physical properties shall comply with requirements of 3.5.2 and 3.5.3.

4.4.1.3 Damaged rotating band (3.5.4). An impression with "cast stone" (or grey stone) shall be taken of each nick, scratch, dent or gouge to determine the depth of the imperfection by a comparator-type device or any other accurate device approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NR4. Any other method devised by the contractor to measure the depth of imperfections shall be approved by Picatinny Arsenal, SMUPA-NR4. For allowable depth of the imperfection see Fig. I. In addition, the rotating band shall be visually inspected for any crack or foreign matter embedded in the rotating band.

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4.4.2 Metal to metal contact (see 3.6).

4.4.2.1 Metal to metal contact (see 3.6.1). Measure the length of the slug to the nearest .001 inch, minimum, mark position and record. Measure distance from forward face to inside shoulder to nearest .001 inch, maximum and mark position and record. Match minimum position of slug with maximum position of distance from face to inside shoulder. Subtract the minimum length of the slug from the maximum length from face to inside shoulder and record. This calculated dimension shall comply with 3.6.1 after assembly of pins when the distance from the face to slug is measured in the marked position.

4.4.2.2 Metal to metal contact (see 3.6.2). The joint between the windshield and body shall be examined with a 10-power magnifier and comply with requirement 3.3.2.

4.4.2.3 Metal to metal contact (see 3.6.3). The gap shall be measured as specified on dwg. 9204730 to insure metal to metal contact. The procedure and equipment used in measuring the gap shall be approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NR4.

4.4.3 Paint adhesion (see 3.8). The test shall be performed on the painted item after a maximum of 24 hours drying. Water resistant, pressure sensitive adhesive tape (3/4 inch width) conforming to the requirements of Federal Specification PPP-T-60, Type II or III, class 1 or 2 shall be used. Press a 2 inch length of a somewhat longer piece firmly onto a flat cylindrical surface of the item, rubbing out all air bubbles under the tape. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape and at a rapid speed strip it from the item by pulling the tape back upon itself at 180 degrees (in such a manner that the tape is folded back to back during the procedure). Observe for bared spots where paint is removed. Disregard flecks of paint on tape where underlying metal or protective finish is not visably exposed.

4.4.4 Charpy and tensile.

4.4.4.1 Charpy and tensile (see dwgs. 9204731, 9204732, 9204733 (tensile not applicable on 9204733), 9214158, 9214159, 4.3.3.2.1, 4.3.3.3.1 and 4.3.3.3.2). The heat-treated lot shall not exceed 25 pieces from the same manufacturer and the same millrun of stock. A sample piece, minimum length one inch, shall be selected from the stock at random when preparing the pieces (see applicable drawing) for heat treatment from stock. This sample may be used for both charpy and tensile specimens. The sample shall be included in the heat-treated lot and receive the same treatment and process as the heat-treated parts. After heat treatment, two transverse charpy specimens, when required, shall be selected from the sample 180° apart, with the notch on one specimen facing the outside of the sample and the notch on the other specimen facing the inside of the sample. Two transverse tensile specimens, when required, shall be selected from the sample 180° apart. When solid stock is used, the specimens shall be selected from a ring area one inch wide from the outside diameter of the sample. The specimens shall be prepared as specified in Federal Test Method Standard No. 151 using methods 211.1 or 221.1 as applicable and tested on approved test equipment. The average of the test results from the two specimens shall comply with drawing requirements.

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4.4.4.2 Charpy or tensile (see dwg. 9203276, 4.3.3.2.2). Two test specimens shall be selected at random from stock for each heat-treated lots of parts. These specimens shall be included in the heat-treated lot and receive the same treatment and process. After heat treatment, the test specimens shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151 using methods 211.1 or 221.1 as applicable. Each test specimen shall comply with drawing requirements.

4.4.5 Shear (see dwg. 9203276 and 4.3.3.4). Two suitable test specimens shall be prepared and tested for each heat-treated lot on approved test equipment. Each specimen shall comply with drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The shell shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The shell shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The shell shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.3 Supersession data. This specification includes the requirements of Purchase Description PA-PD-2584 (Rev. 2) dated 21 March 1966.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1190-A012

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